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Attention-based Spatio-Temporal Graph Convolutional Networks-Enhanced Deep Reinforcement Learning for Adaptive Traffic Signal Control in Urban Traffic Networks

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Abstract

Urban traffic congestion remains a critical challenge due to the inflexibility of conventional fixed-time or rule-based traffic signal control systems, which cannot adapt to real-time dynamic traffic flows. Existing deep reinforcement learning (DRL)-based adaptive signal control methods often neglect complex spatio-temporal dependencies in urban road networks, and graph-based spatio-temporal models (e.g., ASTGCN) are mostly limited to traffic flow prediction rather than signal control optimization. To address these gaps, this study proposes a novel adaptive traffic signal control framework by integrating Attention-based Spatio-Temporal Graph Convolutional Networks (ASTGCN) with Multi-Agent Deep Deterministic Policy Gradient (MADDPG). The ASTGCN module effectively captures spatial topological correlations and temporal dynamic patterns of traffic flow via graph attention and temporal convolution, while the MADDPG framework realizes centralized training and decentralized execution to solve multi-intersection cooperative control under partial observability. Experiments are conducted on the SUMO simulation platform with a four-intersection road network, using both real-world and stochastic traffic flows for validation. Comparative results with fixed-time control, Max-Pressure, centralized/distributed DRL, and vanilla MADDPG baselines show that the proposed method reduces traffic congestion by 20%-25% and achieves superior performance in convergence speed, generalization ability, extreme traffic load resilience, and non-uniform traffic adaptation. Ablation studies further verify the effectiveness of the ASTGCN-based spatio-temporal feature extraction component. This framework provides a scalable and data-driven solution for intelligent traffic signal control in urban traffic networks, supporting the development of smart mobility systems.

Keywords: ASTGCN, Deep Reinforcement Learning, Spatio-Temporal Modeling, Adaptive Traffic Signal Control, Urban Mobility Optimization

Introduction

Traffic congestion in urban networks remains a persistent challenge¹⁻³, exacerbated by the limitations of conventional traffic signal control systems, which rely on fixed cycles or rule-based strategies that fail to adapt to real-time traffic

dynamics. While deep reinforcement learning (DRL) has emerged as a promising tool for adaptive signal optimization, existing approaches often focus on single-intersection scenarios or oversimplify state representations, neglecting critical spatial-temporal interdependencies and practical constraints. Recent studies leveraging graph-based models, such as Spatio-Temporal Graph Convolutional Networks (STGCN)⁴, have demonstrated improved traffic flow prediction by capturing spatial topology and temporal patterns. However, STGCN-based approaches have not yet received significant attention or application in the field of traffic signal control optimization.

Numerous studies have focused on traditional approaches for traffic signal control. For instance, Zoabi R et al.⁵ proposed an enhanced Max-Pressure method that incorporates both bounded and unbounded cycle lengths, which improves fairness in vehicle prioritization and enhances the efficiency of signal timing decisions. Li Q et al.⁶ proposed a spatiotemporal partitioning-based green wave trajectory coordination optimization model, which effectively reduces the average vehicle delay and stop frequency at intersections by synchronizing signal phases across dynamically partitioned traffic corridors. Conventional traffic signal control methods have achieved substantial progress in specific scenarios; however, their generalization performance remains constrained by heavy reliance on prior knowledge, which limits adaptability to dynamic and heterogeneous traffic conditions.

Other approaches have sought to optimize traffic signal control by introducing enhancements to deep reinforcement learning (DRL) across multiple dimensions. For example, Zhang H et al.⁷ proposed a multi-agent sequential decision-making framework that models multi-intersection control as a Markov decision process. This method experimentally demonstrated superior performance over baseline approaches in terms of reducing average vehicle delay and improving intersection coordination efficiency. Nie L et al.⁸ developed a multi-objective reinforcement learning (MORL) framework termed CMRM (Coordinated Multi-Objective Reward Mechanism), which not only optimizes traffic efficiency but also minimizes environmental pollution by dynamically balancing vehicle throughput and emission reduction goals. Duan L et al.⁹ developed a carbon emission-aware reinforcement learning (RL) framework that optimizes intersection sustainability by integrating ecological objectives into state space design. Specifically, experimental results demonstrate decrease in carbon footprint compared to conventional RL baselines, alongside maintained intersection efficiency. Wang J et al.¹⁰ proposed a multi-intersection signal control framework based on asynchronous reinforcement learning, which addresses the heterogeneous impact of rewards across different intersections and accelerates model convergence through decentralized decision-making. Jamil et al.¹¹ proposed a composite reward architecture (CRA) for deep reinforcement learning-based adaptive traffic signal control, which realizes multi-objective optimization via parallel reward calculation and majority voting decision-making. Specifically, the proposed CRA effectively improves traffic performance in terms of vehicle waiting time and queue length compared with traditional reward functions. Nower¹² further developed a dynamic weight-based multi-objective

reward architecture (DWMORA) for adaptive traffic signal control, which adaptively adjusts weight coefficients according to real-time traffic states to overcome the limitation of fixed weights in conventional multi-objective reward functions. Recent advances in reinforcement learning (RL) have demonstrated the versatility of diverse RL approaches in traffic signal control. However, their full potential remains underexplored due to challenges such as partial observability in complex urban networks, scalability limitations in multi-agent coordination, and the absence of standardized reward functions that holistically balance efficiency, fairness, and sustainability.

To address these gaps, this study pioneers a novel framework. The core contributions of this study are clearly presented as follows. Conventionally, Attention-based Spatio-Temporal Graph Convolutional Networks (ASTGCN) have been primarily utilized for traffic flow prediction. However, the innovation of this research lies in its integration of ASTGCN with Multi - Agent Deep Deterministic Policy Gradient (MADDPG), transcending the traditional application scope of ASTGCN. This integration enables centralized training of spatio-temporal traffic features and decentralized signal control across urban networks.

The proposed framework is designed to fully exploit the unique capabilities of ASTGCN in extracting temporal - domain information features, which is another innovative aspect of this method. ASTGCN dynamically weights traffic correlations between adjacent intersections through graph attention layers. This mechanism allows it to capture the complex spatio - temporal dependencies in traffic data more effectively. Meanwhile, MADDPG agents employ prioritized experience replay and policy ensembles to optimize signal phases under partial observability. The combination of these two techniques in the proposed framework holds great potential for improving the efficiency and adaptability of traffic signal control in urban networks.

Case studies further highlight the model's adaptability to varying traffic densities and network topologies, aligning with recent advancements in graph-based traffic forecasting. By bridging the gap between spatial-temporal deep learning and adaptive control, this work provides a scalable, data-driven solution for smart city infrastructure, with potential extensions to multi-agent systems and edge computing deployments.

Problem definition

Consider a transportation network composed of multiple road segments (edges), where each segment contains multiple lanes. Vehicles perform displacement movements within these lanes. Traffic lights, installed at road intersections, govern directional right-of-way permissions through their phase configurations. As illustrated in Figure 1, the network topology in this case study comprises 4 phase-based traffic light groups, 24 road segments, and 72 lanes, with numerous vehicles operating under signal phase control rules.

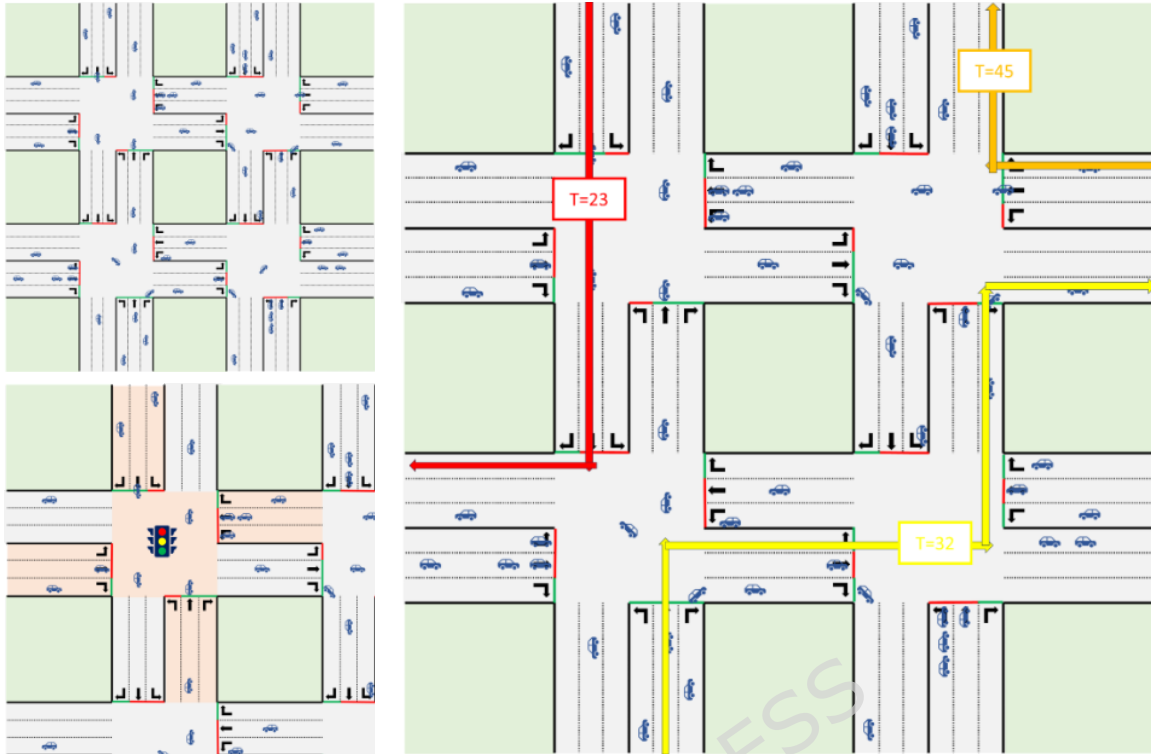


Figure 1 A simple transportation network(upper left), Orange-highlighted lanes indicate the detection scope governed by the top-left traffic light controller(lower left), and Examples of traffic flow: yellow indicates complex multi-directional flow, orange demonstrates simplistic unidirectional movement, and red represents intermediate bidirectional interaction patterns(right).

Intelligent traffic light systems (ITLS), equipped with environmental perception and decision-making capabilities, prioritize conflict-free phase scheduling. Under a right-hand traffic configuration (assumed as the general case), constant right-turn permissions are granted, while left-turn and through movements require phase-segregated conflict avoidance. This study constrains the signal control logic to eight certified conflict-free phases, as systematically defined in Figure 2, which strictly prohibit trajectory overlaps between antagonistic traffic streams.

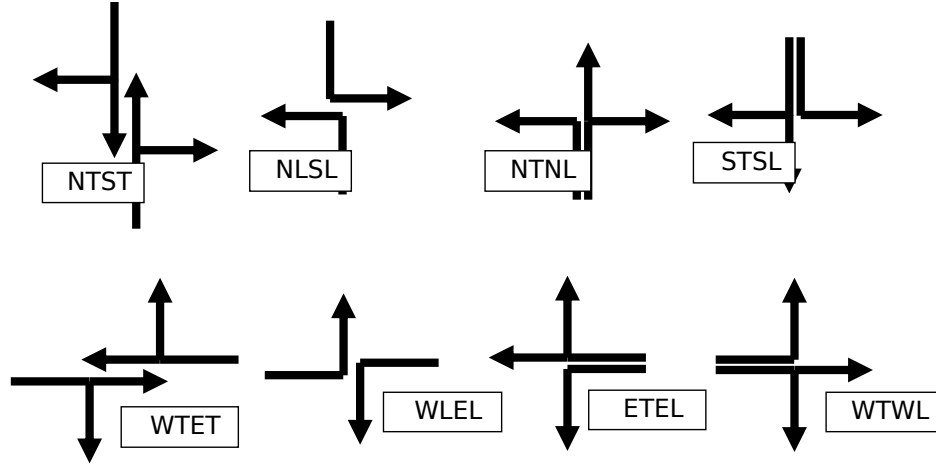


Figure 2 The complete set of available phase combinations

The perception domain of individual traffic signal controllers is architecturally constrained. This study postulates that such controllers monitor all ingress lanes approaching their governed intersections, capturing vehicular metrics including queue density and velocity profiles. Additionally, vehicular states within the central conflict area (CCA) of the intersection are perceptible to the signal devices. Egress lanes exiting the intersection, however, provide no decision-critical data and are therefore excluded from perception parameters. This methodologically bounded perception framework, as schematically delineated in Figure 1, deliberately ignores distal roadway segments beyond the operational zone of influence.

At each decision-making interval, individual traffic signals determine their subsequent phase configuration based on locally observed data. **This optimization process aims to minimize network-wide vehicle stagnation** by ensuring collision-free vehicle movement while maximizing travel efficiency toward destinations.

$$C_{N, F, T} = \sum_{t \in T} \sum_{l_i \in \mathcal{L}} \varpi_{\text{stop}, l_i}(t) \quad (1)$$

Within Equation (1), $\varpi_{\text{stop}, l_i}(t)$ quantifies the total number of queued vehicles across all lanes, where $l_i \in \mathcal{L}$ denotes the i -th lane. The system parameters are defined as follows:

- N : Transportation network topology.
- F : Traffic flow.
- T : Evaluation time horizon.
- $C_{N, F, T}$: Congestion index of the transportation network.

An optimal traffic signal control strategy should prioritize the minimization of $C_{N, F, T}$, thereby reducing systemic delays while maintaining operational safety.

Traffic flow is defined as the temporal sequence of vehicle entry events with predefined trajectory intentions into a transportation network, structured through

connected lane sequences. Within the evaluation time horizon, there coexist vehicles with heterogeneous routing objectives, as exemplified by the flow patterns illustrated in Figure 1. These vehicles maintain adherence to their predefined trajectories while permitting velocity adjustments (acceleration/deceleration) or mandatory stops in response to traffic signal phase transitions.

Traffic signal control strategies must demonstrate robust performance under both stochastically generated traffic flows and empirically validated traffic patterns (derived from real-world roadway data collection).

Method

This section delineates the methodological framework employed in this study, encompassing two principal components: (1) the formalization of a Multi-Agent Markov Decision Process (M-MDP) for signalized intersection control, and (2) the implementation of an Multi-Agent Deep Deterministic Policy Gradient with Attention-based Spatio-Temporal Graph Convolutional Networks (ASTGCN-MADDPG) algorithm for decentralized traffic signal coordination.

Multi-Agent Markov Decision Process

M-MDP is a formal framework for sequential decision-making in stochastic, cooperative, or competitive environments where multiple agents interact under partial observability. It extends the classical POMDP model to multi-agent systems and is defined by the 8 tuples:

$$\langle N, S, \{A_i\}, T, \{R_i\}, \mathcal{F}, \{O_i\}, \gamma \rangle$$

N represents that there are N agents in the Markov model. S represents the finite set of global system states, which cannot be directly observed by any individual agent. $\{A_i\}$ denotes the joint action space formed by the Cartesian product of individual agents' action sets $A_1 \times A_2 \times \dots \times A_N$, where each A_i corresponds to the executable actions of agent i . $T: S \times \{A_i\} \times S \rightarrow [0, 1]$ specifies the state transition probability function, determining the likelihood of transitioning to a new state given the current state and joint actions. $\{R_i: S \times \{A_i\} \rightarrow \mathbb{R}\}$ defines the reward function for each agent, which may align (cooperative) or conflict (competitive) with others' objectives. $\{O_i\} = \mathcal{F}(S)$ defines the observation space of each agent with observation rules $\mathcal{F}$. $\gamma \in [0, 1)$ is the discount factor for cumulative reward.

Traffic signal control can be effectively modeled as a M-MDP due to its inherent sequential decision-making characteristics. The modeling method of our research is described below:

Observation space $\{O_i\}$: the observation space O_i of traffic intersection I_i is defined as follows:

$$o_{i,t} \in O_i, o_{i,t} = [n_{\text{lane}, I_i, t}, n_{\text{stop}, I_i, t}, v_{\text{average}, I_i, t}] \parallel I_i, t' = \{t, t-1, t-2\}$$

L_i is the observation range of signal light l_i shown in Figure 1. $n_{lane, l, t} \in \mathbb{N}^{L_i/}$ is a cardinality vector enumerating vehicular presence per traffic lane. $n_{stop, l, t} \in \mathbb{N}^{L_i/}$ is a stationary vehicle vector quantifying queuing behavior, representing the number of vehicles exhibiting zero-velocity states ($v \leq v_{stop}$) in lane l . $v_{average, l, t} \in \mathbb{R}^{L_i/}$ is a kinematic state vector denoting the space-mean speed of vehicular flow in lane l , calculated through temporal aggregation of individual vehicle speeds within the detection region. The Markov state space incorporates observations from three consecutive timesteps $\{t, t-1, t-2\}$ designed to capture temporal dependencies that enhance feature extraction in subsequent reinforcement learning (RL) processes.

Each traffic light can observe all approaching roadways converging into the intersection, including the central zone, which is conceptualized as a distinct lane type. In typical bidirectional X-lane road networks' intersection, $|L_i|$ could be computed by:

$$|L_i| = 4 * X + 1$$

Action space $\{A_i\}$: without loss of generality, we consider the right-hand traffic system as an illustrative case. The traffic signal phase configuration for a specific movement direction exhibits three operational states: 1) red light prohibition, 2) permitted left-turn movement, and 3) permitted right-turn movement and straight. Through combinatorial analysis of these mutually exclusive states while maintaining traffic conflict avoidance principles, the system yields eight distinct non-conflicting phase combinations that satisfy vehicular movement compatibility requirements, as illustrated in Fig. 2, representing the complete set of available phase combinations. This means that the action space of each agent is discrete, with a capacity of 8.

Agent number N : the parameter N , representing the total agent population, is determined by the topological configuration of the transportation network and remains invariant during simulation periods. Each autonomous agent is capable of perceiving localized environmental states $\square_i \in O_i$ within its observable radius and executing corresponding control actions $a_i \in A_i$ based on real-time feedback mechanisms.

State transition probability function T : Under green signal phase with absence of preceding vehicles, the vehicle will accelerate to the maximum allowable speed. Otherwise, the changes of vehicle target speed on the traffic network meet car-following model shown in formula 2, which forms collision-free operation:

$$v_{safe} = v_{leader} + \frac{gap - l_{veh}}{\tau + \frac{v}{b_{max}}} (2)$$

In this formulation, the variable v_{leader} denotes the instantaneous velocity of the leading vehicle, gap quantifies the dynamic inter-vehicle spacing, and τ corresponds to the human operator's neuro-muscular latency. The parameter v represents the ego vehicle's real-time kinematic state, while b_{max} defines the physical deceleration threshold constrained by the vehicle's

braking system dynamics. When a vehicle encounters a red light, it is considered to be following a stationary vehicle on the stop line.

Reward $\{r_i\}$: to minimize congestion $C_{N, F, T}$, the reward for each agent is calculated in formula 3:

$$r_i(t) = - \sum_{l \in \mathcal{L}_{l_i}} [\square_{\text{stop}, l}(t) + \square_{\text{lane}, l}(t)](2)$$

$\mathcal{L}_{l_i}$ represents the set of lanes that can be observed by intersection l_i . The more vehicles around an intersection, the more severe the reward/punishment, with the penalty for stationary vehicles being twice that for moving vehicles by the experiences that it is often stationary vehicles that cause traffic congestion. (The stationary vehicle penalty is concurrently computed for both $\square_{\text{stop}, l}$ and $\square_{\text{lane}, l}$.)

Multi-Agent Deep Deterministic Policy Gradient

The Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm is a cutting-edge reinforcement learning framework designed to address the complexities of multi-agent environments, particularly in mixed cooperative-competitive scenarios. Built upon the actor-critic architecture, MADDPG adopts a "centralized training with decentralized execution" paradigm to resolve environmental non-stationarity and high variance challenges inherent in multi-agent systems.

In our problem, each agent maintains an actor network, while the traffic network maintains a critic network. Actor networks generate deterministic policies $\pi_{i\theta}$ based on local observation o_i . The policy π_θ is a mapping from state to action:

$$a_i = \pi_{i\theta}(o_i), i = 1, 2, \dots, N$$

The critic network evaluates joint action-value functions using global state information and all agents' actions during training. This centralized critic design stabilizes learning by mitigating observational partiality while preserving execution-time autonomy.

In our problem, the critic network generates the joint action value of the current state, denoted as $Q(s, a_1, a_2, \dots, a_N)$. During training, the critic network leverages global information to stabilize learning, while actors rely solely on local observations. Agents interact with the environment using their current policies $\pi_{i\theta}$ and noise exploration. Transitions (o_i, a_i, r_i, o'_i) are stored in a replay buffer D , ensuring decorrelation of training data.

The critic minimizes the temporal difference (TD) error using a batch of sampled experiences:

$$\mathcal{L}(Q_i) = E_D[(Q_i(s, a_1, a_2, \dots, a_N) - y_i)^2]$$

$$y_i = r_i + \gamma Q_i(s', \pi_{i\theta}(o'_1), \pi_{i\theta}(o'_2), \dots, \pi_{i\theta}(o'_N))$$

where the prime notation (s' or o') denotes the subsequent state/observation in the transition process. $\mathcal{L}(Q_i)$ is used to update Q_i to ensure training stability via soft updates

$$Q_i' = \tau Q_i + (1 - \tau) Q_i.$$

The actor is updated using the deterministic policy gradient:

$$\nabla_{\theta} J(\pi_{\theta}) = E_D[\nabla_{\theta} \log \pi_{\theta}(o_i) Q_i(s, a_1, a_2, \dots, a_N)]$$

This gradient directs the actor to maximize the critic's estimated Q-value for its actions. After updating the actor network and critic network in turn, a

strategy can be generated to maximize rewards. Considering that $\square_{\text{lane}, i}(t) - \square_{\text{stop}, i}(t)$ is inherently dependent on the policy-independent traffic flow, the minimization of $C_{N, F, T}$ will be achieved when the Markovian reward is maximized, thereby resulting in the mitigation of road congestion.

Attention-based Spatio-Temporal Graph Convolutional Networks

The Attention-based Spatial-Temporal Graph Convolutional Network (ASTGCN) is a novel deep learning architecture designed for traffic flow forecasting, addressing the complex spatiotemporal dependencies inherent in transportation systems. In this work, we innovatively employ ASTGCN as the core feature extractor of the MADDPG actor network, which is the critical rationale for integrating ASTGCN with MADDPG. By integrating graph convolution with attention mechanisms, ASTGCN dynamically captures both spatial correlations between road network nodes and temporal patterns across historical sequences. This dual-attention mechanism enables adaptive focus on dynamically changing traffic patterns, while the multi-component architecture effectively handles heterogeneous periodic features in urban mobility. Compared with spatial-only GNN (which merely models topological correlations) and vanilla MADDPG (which lacks spatiotemporal feature extraction), ASTGCN provides complete spatiotemporal representations of traffic states, which is essential for multi-intersection cooperative signal control.

Notably, although primarily applied to traffic flow prediction, ASTGCN's ability to model spatiotemporal traffic dynamics makes it inherently suitable for traffic signal control—a task fundamentally governed by real-time flow characteristics. The end-to-end spatiotemporal modeling of ASTGCN is tightly coupled with the centralized training/decentralized execution paradigm of MADDPG, which enhances the perception and decision-making capabilities of multi-agent systems. The architecture's end-to-end learning framework could potentially optimize signal timing strategies by directly mapping observed traffic states to control actions, bypassing traditional rule-based heuristics.

Our study integrates the ASTGCN as the policy network (actor) within the MADDPG framework. Specifically, given that the input to the actor network consists of intersection perception data structured as a 4D tensor [batch_size, c_in, ts, n_vector], the network processes this spatiotemporal representation and generates an 8-class categorical output. This output represents the optimal traffic signal phase to be implemented at the intersection under the current policy, where each class corresponds to a distinct phase configuration in the signal control scheme.

$$a = \text{FC}(\text{GraphConv}(\text{TemporalConv}(x)))$$

The TemporalConv module, a GLU-based (Gated Linear Unit) temporal convolution block, comprises a temporal convolutional network (TCN), residual connections, and activation functions, as illustrated in Fig. 5. This architecture is specifically designed to capture temporal dynamics at intersections through hierarchical feature abstraction. By processing sequential traffic patterns with dilated causal convolutions, the module enables the identification of real-time traffic flow trends (e.g., sudden

congestion buildup or dissipation), thereby enhancing spatiotemporal feature extraction capabilities through its gated mechanism. The residual connections further stabilize gradient propagation during long-term dependency learning, which is critical for maintaining temporal coherence in traffic state transitions.

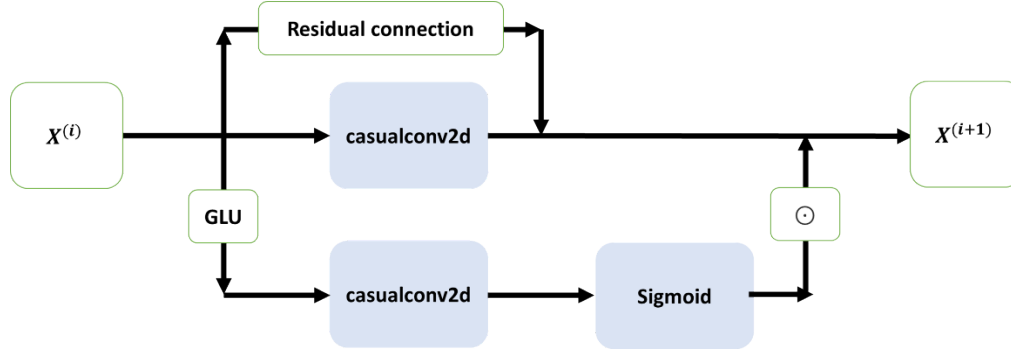


Figure 3 The structure of TemporalConv

After temporal features are extracted by TemporalConv, the resulting features are then fed into Chebyshev Graph Convolutional (ChebGraphConv) layers for spatial feature extraction. Within this graph construction framework, each traffic lane is modeled as a node in the graph data structure, while the connectivities between lanes are represented as edges. The directional properties of traffic lanes are integrated by modeling the directional flow as edge orientations in the directed graph. Thus, this integrated approach forms a comprehensive graph representation that encodes the current state of lane network topology, as depicted in Figure 4.

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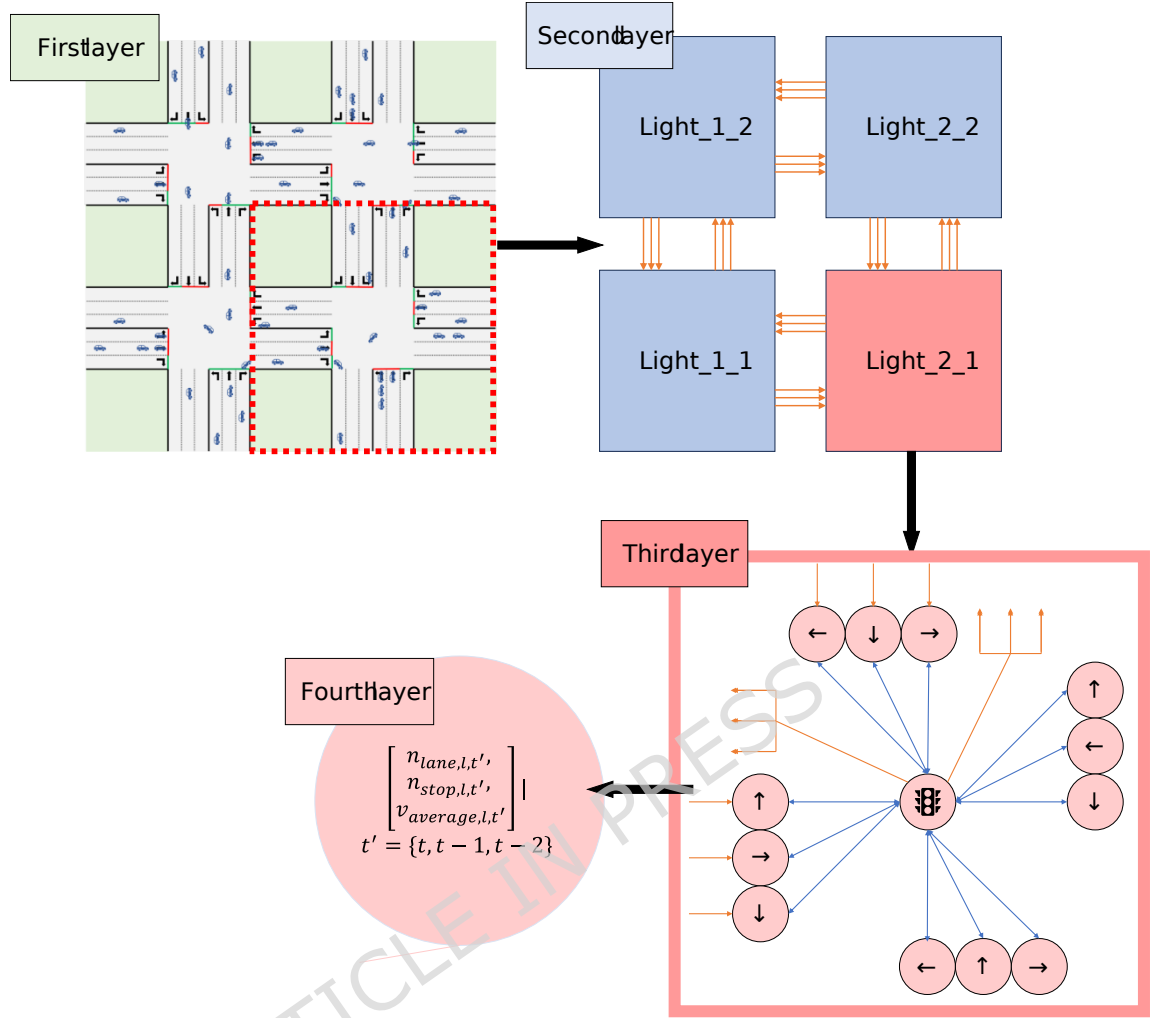


Figure 4 The graph data structure of the transportation network shown in Figure 1. First layer: The outermost transportation road network, encompassing all traffic signals and road vehicle information, with direct integration into the simulation. Second layer: The graph structure model of the transportation road network, serving centralized reinforcement learning training for the critic network. Here, each traffic signal has a limited decision-making field of view; during training, only limited information can be transmitted with adjacent signals, and decision-making is confined solely to the signal's own field of view. Third layer: The graph data structure for the decision-making field of view of a single traffic signal, specifically targeting the red signal illustrated in the second layer. When training the critic network, both the orange directed edges of graph nodes (cross-field-of-view information, i.e., information exchange with the blue signal) and blue directed edges (information within the red signal's field of view) are utilized. In contrast, only the blue directed edges are employed when the actor network executes decision-making. Fourth layer: Information of an individual graph node, which can fall into one of four categories: left-turn lane, right-turn lane, straight lane, or intersection center. Each node's information includes current road vehicles, vehicles stopped on the road, average vehicle speed, and historical data from the two preceding time steps.

During the reinforcement learning process, the actor network employs only the subgraph depicted in Figure 4 when making decisions. For example, the northeast-directional traffic lights exclusively use the red subgraph in the upper-left corner of Figure 4, thus excluding the observation scopes of other traffic signals (blue lights in Figure 4). In contrast, the critic network incorporates the complete graph for action-value estimation during its evaluation phase. This design ensures that the model benefits from global information guidance throughout MADDPG training, leading to improved policy quality. Meanwhile, during policy execution, computational costs are reduced via localized subgraph processing, enabling real-time policy generation while preserving decision-making efficiency.

To extract spatial features, the graph data is fed into the *ChebGraphConv* module. *ChebGraphConv* leverages Chebyshev polynomial approximation to perform localized spectral filtering on graph-structured data. Unlike traditional spectral convolution requiring explicit eigen decomposition of the graph Laplacian matrix, *ChebGraphConv* approximates the spectral filter using K -th order truncated Chebyshev polynomials. This avoids direct eigenvalue decomposition and reduces computational overhead while capturing multi-hop neighborhood information.

The convolution operation is defined as:

$$g_{\theta}(L) \cdot X = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{L})X$$

where θ_k is learnable polynomial coefficients. $T_k(\tilde{L})$ is k -th order Chebyshev polynomial evaluated at $\tilde{L}$, recursively computed as:

$$T_0(\tilde{L}) = I, T_1(\tilde{L}) = \tilde{L}, T_k(\tilde{L}) = 2\tilde{L}T_{k-1}(\tilde{L}) - T_{k-2}(\tilde{L})$$

$\tilde{L}$ is the normalized Laplacian matrix, which is determined by the graph structure.

$$\tilde{L} = \frac{2}{\lambda_{\max}}(I - D^{-1/2}AD^{-1/2}) - I$$

Where A is the adjacency matrix, D is the degree matrix, and $\lambda_{\max}$ is the maximum eigenvalue of the Laplacian matrix.

After being processed by the *TemporalConv* and *ChebGraphConv* layers, the model derives spatio-temporal features of the current state. These features are then fed into a fully connected layer. For the actor network, this stage yields the action to be taken according to the current policy, while the critic network estimates the value function associated with the current state. The overall MADDPG architecture with *TemporalConv* and *ChebGraphConv* is illustrated in figure 5.

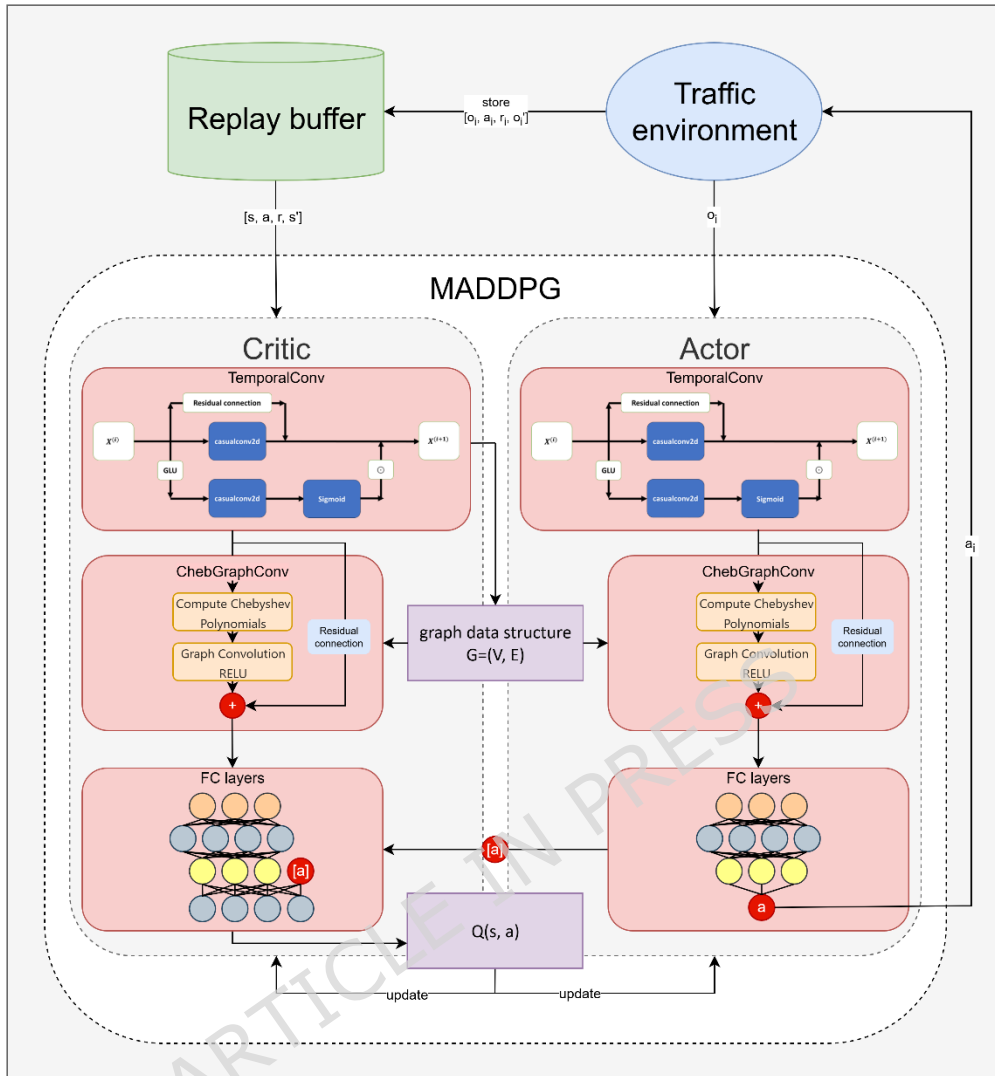


Figure 5 MADDPG architecture with TemporalConv and ChebGraphConv

The computational framework of the proposed method is formally presented in Algorithm 1.

Algorithm 1 MADDPG with TemporalConv and ChebGraphConv

Input: traffic network $N = (J, \mathcal{E})$, traffic flow $\mathcal{F}$

Initialize: actor $\pi_{i\theta}$ and critic Q_i with TemporalConv and ChebGraphConv, replay buffer D

for episode = 1 **to** max_training_episodes **do****for agent = 1 to N do****for** t = 1 **to** max_episode_length **do**

Get current observation $[o_1, o_2, \dots, o_n]$ and state s

$$a_i = \pi_{i\theta}(o_i)$$
$$(r_1, r_2, \dots, r_n), o' = T(s, a_1, a_2, \dots, a_n)$$

Add ($a_1, a_2, \dots, a_n, r, o'$) **to** D

```
if size(D) > batch_size do
```

```

Sample random minibatch from D
Update  $\theta$  via:
 $\mathcal{L}(Q_i) = E_D[(Q_i(s, a_1, a_2, \dots, a_N) - y_i)^2]$ 
 $y_i = r_i + \gamma Q_i(s', \pi_{i\theta}(o'_1), \pi_{i\theta}(o'_2), \dots, \pi_{i\theta}(o'_N))$ 
 $\nabla_{\theta} J(\pi_{\theta}) = E_D[\nabla_{\theta} \log \pi_{\theta}(o_i) Q_i(s, a_1, a_2, \dots, a_N)]$ 
    end for
  end for
end for
return  $\pi_{i\theta}, i = 1, 2, \dots, N$ 

```

Experience

This section elaborates on the experimental setup of the proposed methodology and outlines the established baseline methods used for comparative analysis.

Experiment Setup

The proposed methodology utilizes the Simulation of Urban Mobility (SUMO) platform, an open-source microscopic traffic simulation suite developed by the German Aerospace Center (DLR) starting in 2001. As a continuous-time, multi-modal simulator, SUMO enables high-fidelity modeling of vehicular dynamics, pedestrian movements, and public transport systems via its discrete-time spatial continuum framework.

The experimental road network adopted in this study consists of four interconnected intersections, with each arterial roadway equipped with three directional lanes (as illustrated in Fig. 8).

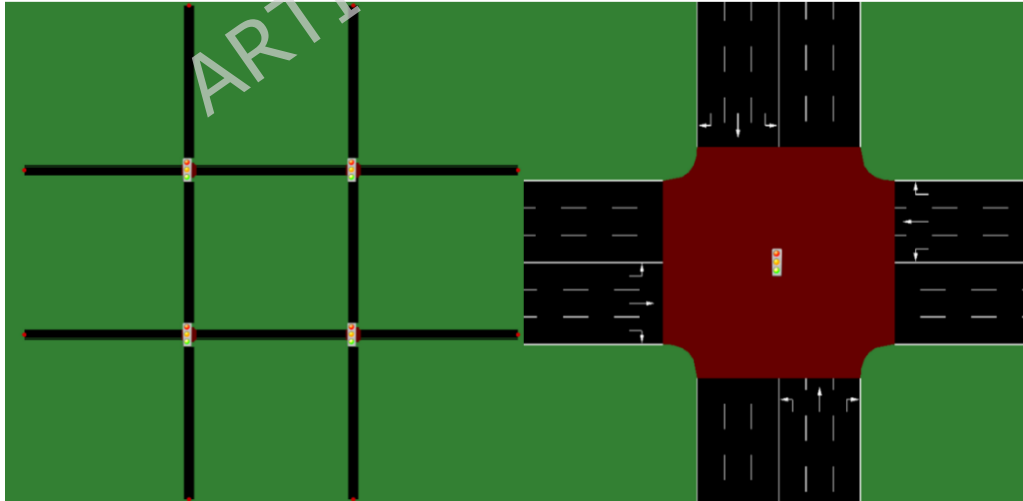


Figure 6 experimental road network

The traffic flow generation methods utilized in the experiments encompass two modes: random traffic flow and traffic flow extracted from real-world traffic data. In this study, both types of traffic flows will be used to validate

the generalization capability of the proposed method and to compare it against baseline methods under a standardized experimental environment.

The actual traffic flows employed in this experiment are derived from <https://traffic-signal-control.github.io/dataset.html>, comprising a publicly accessible Hangzhou traffic network dataset and a Gaussian-sampled traffic flow dataset extended from real-world measurements. These traffic flow datasets retain consistent patterns within individual experiments while exhibiting diverse distributional characteristics. For instance, the traffic flow patterns depicted in Figure 1 may also occur in real-world scenarios.

Random traffic flows fluctuate across each experimental run, with each flow type assumed to follow a Poisson distribution within the traffic network. The parameters of the Poisson distribution are obtained from statistical analyses of real-world traffic datasets. This time-varying traffic flow is employed to evaluate the generalization capability of each method, given that actual traffic flows exhibit similar variability.

Baseline Methodologies

This section elaborates on the baseline methodologies utilized for comparative analysis in this study.

- a) **Fixed-Time Control** [13]: As the foundational approach in traffic signal optimization, fixed-time control employs predefined phase sequences with fixed green durations. Derived from historical traffic patterns using methods such as Webster's formula, this approach calculates phase durations via the formula:

$$C = \frac{1.5L + 5}{1 - Y}$$

where C denotes cycle length, L represents total lost time, and Y is the sum of critical flow ratios. The baseline signal timing scheme was configured with a total cycle length C of 90 seconds, consisting of 33-second green phases for through traffic and 6-second green phases for left-turn traffic, with phase transitions incorporated via a 3-second yellow light interval.

- b) **Max-Pressure Control** [14]: This queue-theoretic approach dynamically prioritizes signal phases with maximum pressure differentials between inflow and outflow lanes. Given that the maximum saturation flow rates are uniform across all directions in this experiment, the pressure metric is formalized as:

$$P_i = \sum_L n_{\text{stop}}$$

By activating phases with maximal P_i , the method mitigates intersection imbalance.

- c) **Distributed Reinforcement Learning** [15]: Multi-agent DRL frameworks facilitate decentralized optimization via localized observations and coordinated policies. Under this configuration, **individual intersections** undergo localized training by **exclusively leveraging local data streams**, thereby **eliminating inter-junction communication dependencies** while maintaining operational self-sufficiency. As a baseline method, PPO reinforcement learning is employed to formulate policies for each intersection.
- d) **Centralized Reinforcement Learning** [16]: In contrast to distributed strategies, the centralized approach integrates observation spaces into a unified state representation, facilitating **single-agent reinforcement learning** within a centrally coordinated optimization framework. As a baseline method, PPO reinforcement learning is utilized to formulate policies for the entire network.
- e) **MADDPG without Graph Neural Networks** [17]: An observation feature extraction network lacking graph neural network (GNN) architectures was implemented, which involves storing observed data as vector-structured representations instead of graph-structured embeddings. This configuration was systematically validated through ablation studies to evaluate the effect of graph-based processing in the proposed methodology.
- f) **MADDPG with Graph Neural Networks**: This experiment involves exclusively the GNN component and **excludes time-domain information extraction**. It is designed to perform an ablation study on the time-domain design of the proposed method, aiming to evaluate the effectiveness of time-domain feature extraction.

Together, these two settings (e) and (f) form a systematic ablation study to validate the necessity of graph spatial modeling and temporal feature extraction, respectively, when integrated with MADDPG.

Experiment Setup

This study systematically evaluates the performance of multiple intersection signal control strategies through a series of experiments specifically designed to assess their robustness, generalization capability, and adaptability under varied traffic conditions. The experimental framework and corresponding performance metrics are elaborated in the following sections.

- a) **Average Reward Across Multiple Trials**: To ensure statistical reliability, each signal control strategy is tested under uniform stochastic conditions across repeated trials. The **average cumulative reward—which represents the overall traffic congestion reduction—is calculated as the primary performance metric**. This approach mitigates the influence of random traffic flow fluctuations and provides a standardized framework for strategy comparison.

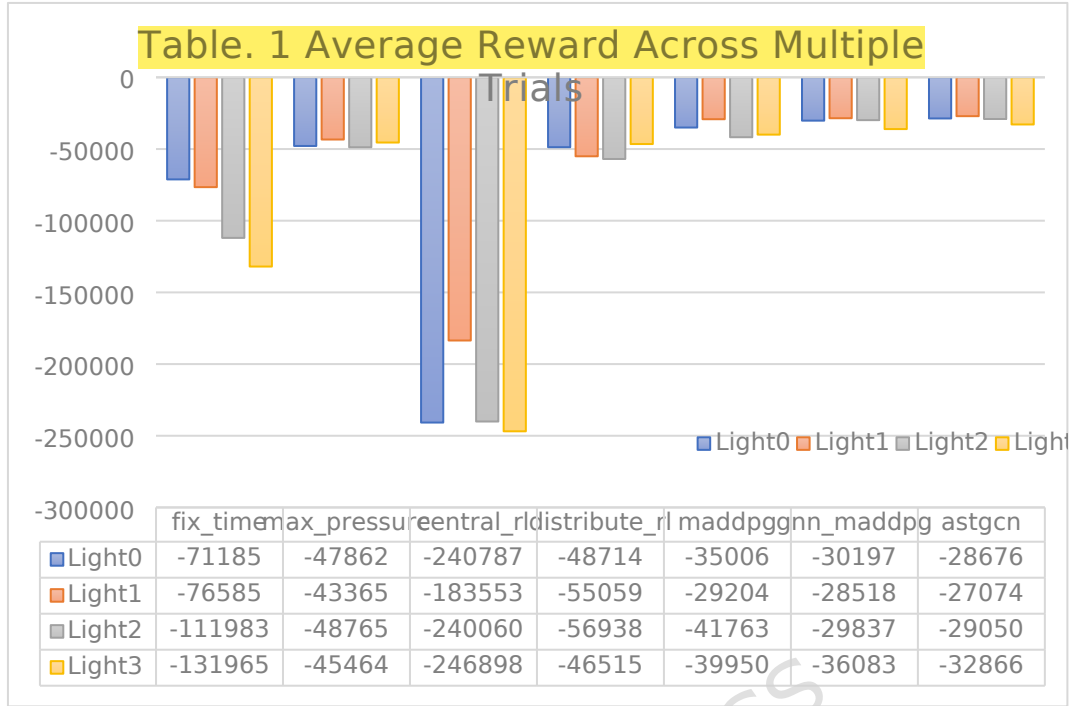
- b) **Generalization Performance**: The generalization capability of each strategy is evaluated by systematically varying traffic flow rates while keeping control policies constant. This test simulates real-world demand variability, e.g., unexpected surges during peak hours or seasonal fluctuations. The mean reward deviation across diverse randomized scenarios is computed to quantify robustness. Strategies showing low deviation are considered more reliable for deployment in dynamic environments. Additionally, cross-validation is performed by training policies on one dataset and testing on another with distinct traffic flow distributions.
- c) **Extreme Traffic Load Resilience**: Strategies are subjected to $2\times$ and $3\times$ traffic volume amplifications to simulate extreme scenarios (e.g., major public events or infrastructure failures). Performance is measured using a degradation ratio that quantifies the relative decline in average reward under extreme loads versus baseline conditions.
- d) **Non-Uniform Traffic Flow Adaptation**: To address anisotropic traffic patterns (e.g., dominant directional flows during rush hours), the experiments introduce asymmetric vehicle arrival rates at different intersection approaches. Performance is evaluated using a directional fairness index that describes the variance in waiting times between high- and low-traffic lanes.
- e) **Convergence Analysis of Reinforcement Learning Methods**: For RL-based strategies, two critical training properties are investigated: convergence speed and convergence stability. The former refers to the number of training episodes required to achieve stable policy performance, while the latter is defined as the variance in rewards during the final 10% of training episodes, which indicates robustness to hyperparameter settings.

Results

Through multiple comparative experiments between this method and the baseline method, the degree of traffic congestion induced by each method is evaluated under identical random number conditions, with the results presented as follows.

a) Average Reward Across Multiple Trials

The comparative performance metrics of the evaluated methods are presented in Table. 1. As evidenced by the data, the Centralized Reinforcement Learning method demonstrates the lowest operational efficacy. This can be attributed to the fact that under the same number of training iterations, the excessively large state space of centralized reinforcement learning hinders its ability to effectively converge to a dominant strategy. The fix-time method also exhibits suboptimal performance, primarily due to its inability to dynamically adapt to traffic flow fluctuations. This rigidity results in inefficient traffic light cycles where numerous vehicles experience unnecessary idling during red phases, consequently generating excessive penalty accumulations.



As outlined in the comparative analysis, the max-pressure method and decentralized reinforcement learning approach exhibit statistically similar performance outcomes, both achieving near-optimal thresholds within the defined evaluation parameters. However, this methodological framework is constrained by inherent architectural limitations rooted in inadequate multi-agent coordination mechanisms.

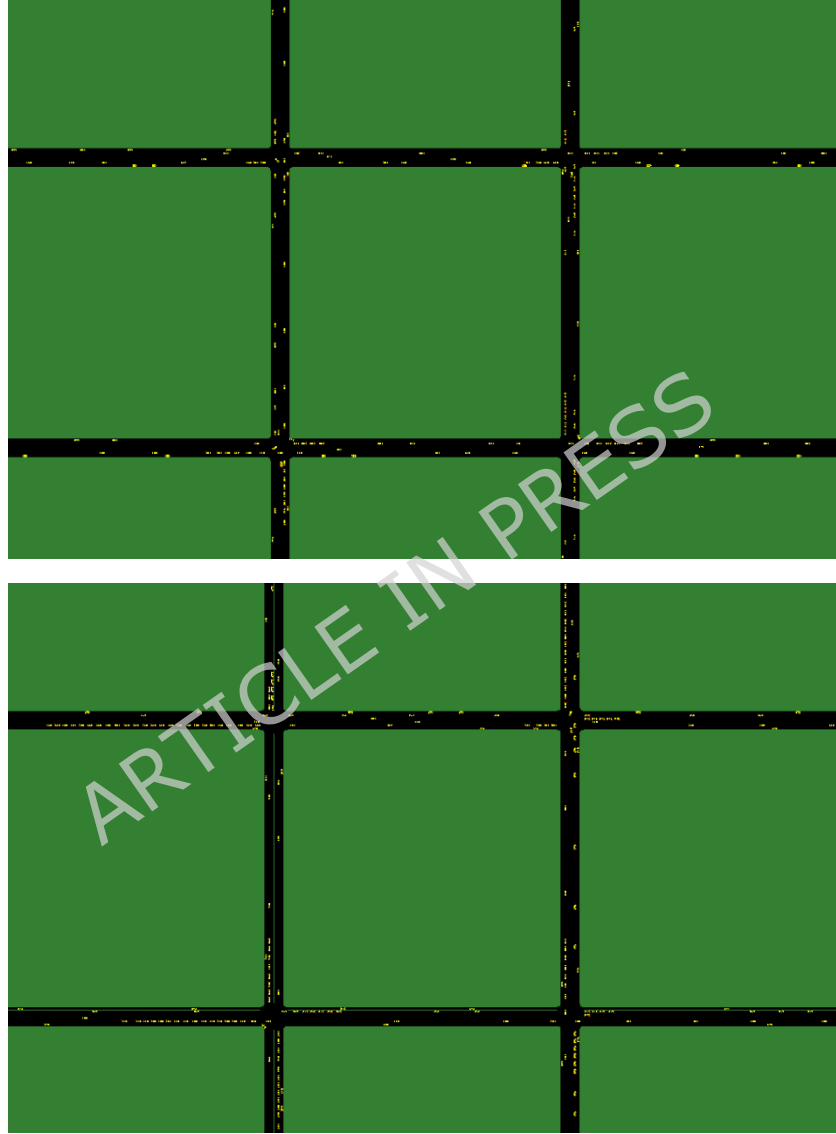
The fully autonomous operation of individual intersection controllers creates informational silos, thereby precluding the exploitation of critical topological features—including but not limited to roadway connectivity patterns and network-wide congestion propagation dynamics. Consequently, the absence of cooperative decision-making frameworks leads to suboptimal global traffic state optimization, particularly in scenarios requiring holistic adaptation to traffic network dynamics.

Explicit ablation studies corresponding to baseline settings (e) and (f) are conducted to verify the effectiveness of each key component in our framework. The MADDPG framework outperforms baseline methodologies in key performance indicators, with its operational efficacy primarily attributed to the synergistic integration of centralized training and distributed execution paradigms. This architectural innovation effectively addresses the exploration-exploitation trade-off inherent in multi-agent adversarial learning environments, while maintaining computational tractability for real-time traffic signal control.

Notably, the ablation experiments clearly validate the necessity of spatio-temporal feature extraction that integrating the ASTGCN for spatio-temporal feature extraction empowers intersections to fully leverage information embedded within the state space, thereby enabling more optimal decision-making. Compared to the baseline MADDPG-only approach, the incorporation of a GNN module reduces

traffic congestion by 15%, while the additional integration of ASTGCN yields a 20% reduction in congestion levels, which strongly demonstrates the superiority of the proposed ASTGCN-MADDPG methodology.

Figure 7 intuitively demonstrates the performance comparison between the ASTGCN+MADDPG method and the direct application of the MADDPG method. It can be observed that the latter exhibits significantly more severe congestion.

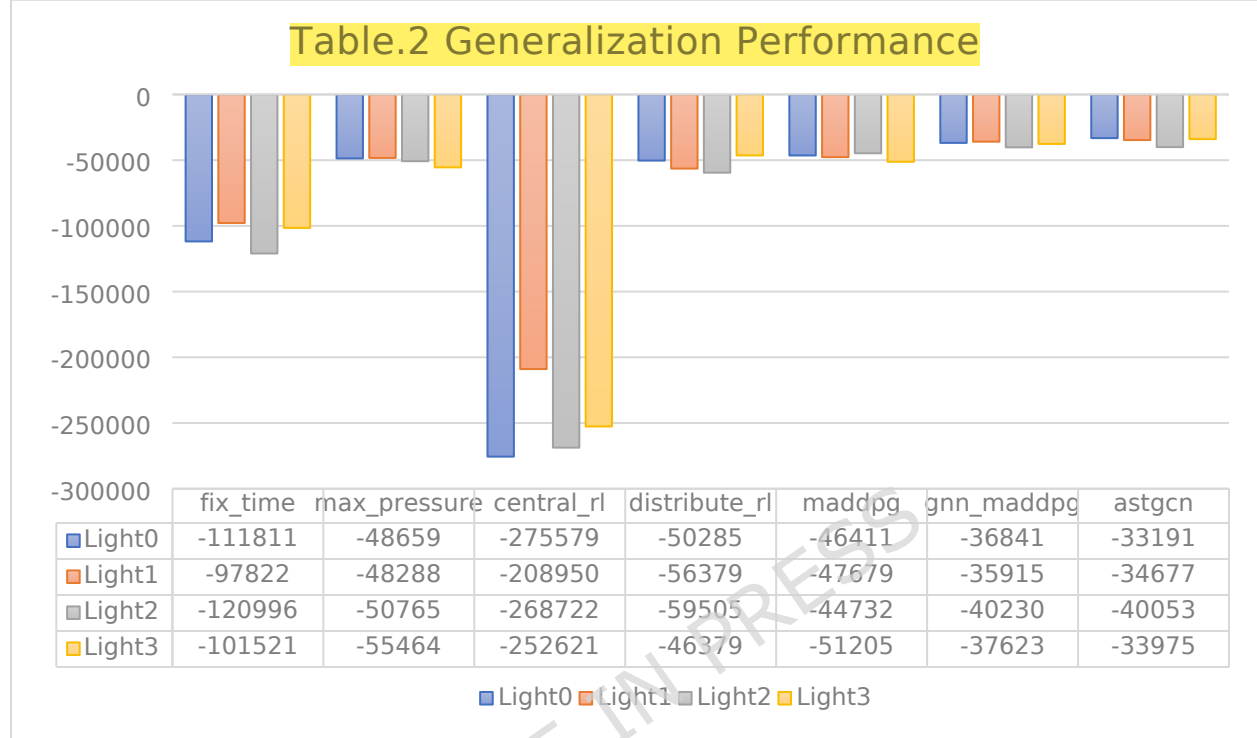


b) Generalization Performance

Figure 7: Schematic diagrams of the ASTGCN+MADDPG intersection road network (upper) and MADDPG alone (lower)

To systematically assess cross-scenario adaptability, we employed a stochastic variation framework wherein traffic flow generation parameters deviate from stationary Poisson assumptions via time-dependent perturbations. Specifically, a non-stationary traffic regime was established by introducing Gaussian-distributed parameter fluctuations ($\sigma = 10\%$ of the mean arrival rate) at each time interval,

emulating realistic demand volatility patterns. Under this intentionally designed stress-testing environment, all benchmark algorithms underwent comparative robustness analyses using Monte Carlo simulations with 1,000 stochastic realizations. The experimental findings are documented in Table 2.



Integrating the findings of both experiments reveals that when stochastic traffic flows are introduced, the performance of all methods deteriorates. Notably, the method proposed in this paper exhibits the smallest performance degradation among multi-agent reinforcement learning approaches. Specifically, the reinforcement learning method combined with GNN reduces traffic congestion by 20% compared to the general MADDPG method, while the approach integrating ASTGCN achieves a 25% reduction in congestion levels. These results demonstrate that the proposed method possesses superior generalization capability.

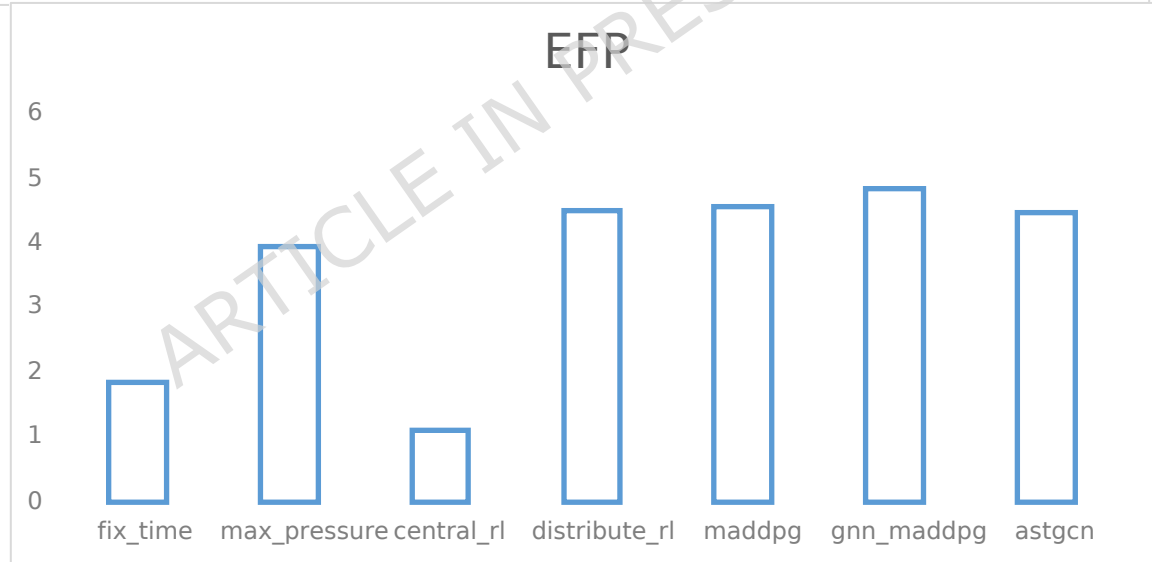
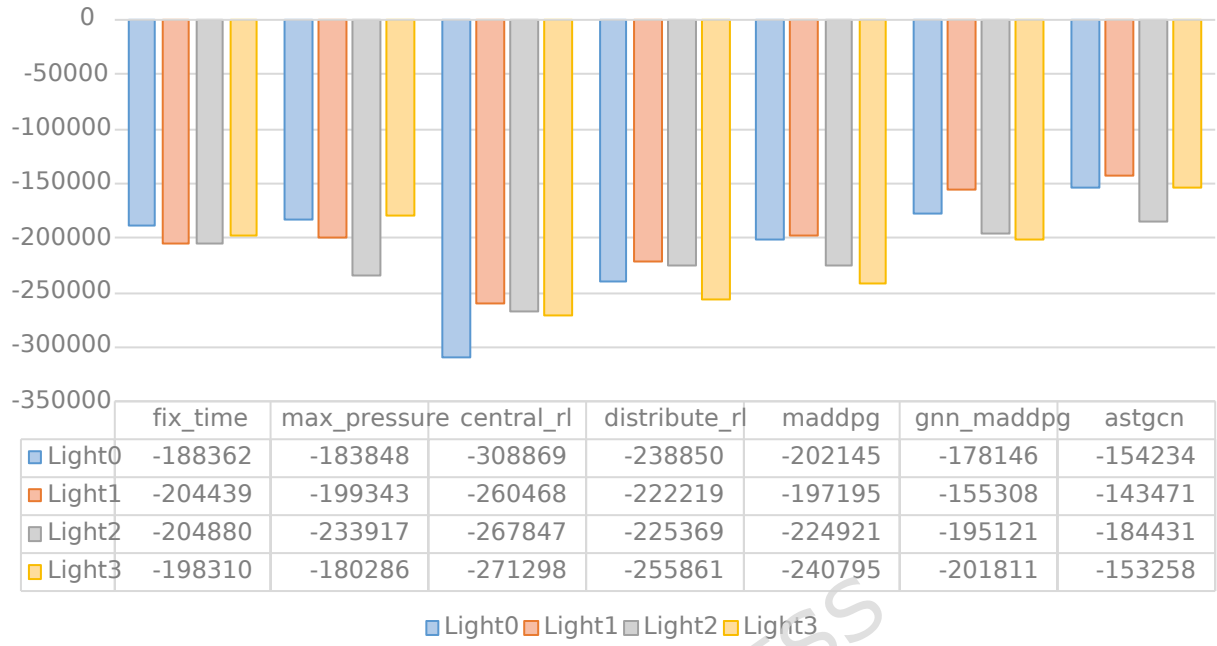
c) Extreme Traffic Load Resilience

The Extreme Flow Penalty (EFP) serves as a metric to quantify the methodology's generalization capability under extreme traffic scenarios.

$$EFP = \frac{r_N - N * r_1}{r_N} * 100\%$$

This parameter is operationally characterized through comparative return analysis, where r_1 represents the baseline return under normal traffic conditions, r_N corresponds to the return generated under N-fold amplified traffic volume.

Table. Extreme Traffic Load Resilience

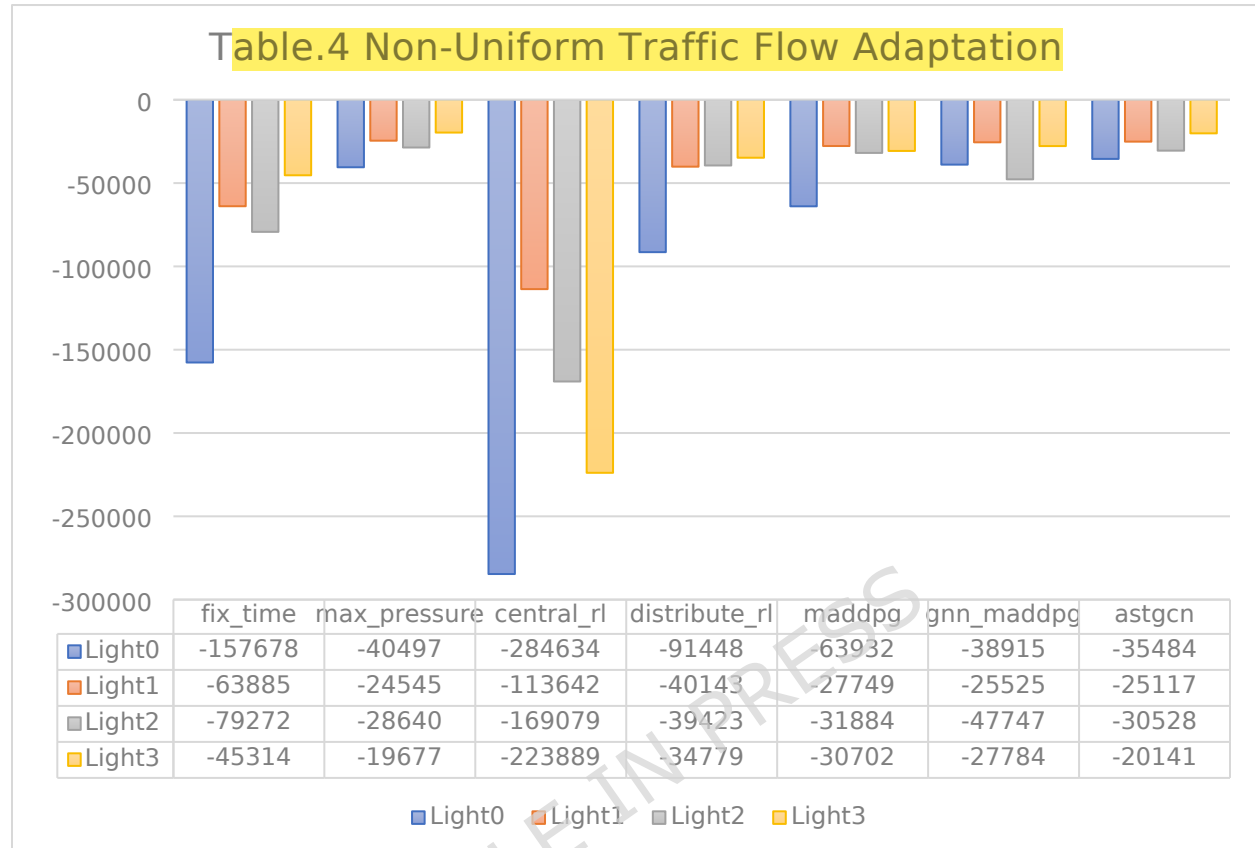


The result of doubling the traffic flow in the road is shown in Table 3. Due to the fact that the basic traffic flow is already extremely high, the doubled traffic flow causes the fix_time method and the centralized reinforcement learning method to fall into severe congestion. Among the other methods that are not yet congested, the method proposed in this paper still achieves the best effect and demonstrates the best EFP performance among all multi-agent reinforcement learning methods.

d) Non-Uniform Traffic Flow Adaptation

To generate a heterogeneous flow distribution with the aggregate traffic volume kept constant, we randomly selected half of the traffic streams to be invalidated and doubled the remaining streams. As shown in Table 4, the experimental results

under this reconfigured environment demonstrate significant performance differences among the evaluated methods.



It is evident from the experiment that the traffic flow around Light0 contains significantly more vehicles than other locations, a scenario where the max-pressure algorithm typically gains an advantage. However, the proposed method not only outperforms other reinforcement learning approaches but also surpasses the max-pressure algorithm itself. By enabling coordinated decision-making across intersections, the ASTGCN+MADDPG method efficiently manages the heavy traffic near Light0, leading to superior overall performance. These experimental results fully validate the robustness of the proposed method in addressing extreme and sudden traffic conditions.

e) Convergence Analysis for Reinforcement Learning Methods

The convergence behaviors of various reinforcement learning algorithms are compared and visualized via line charts, with their quantitative evolution patterns systematically presented in Figs. 10.

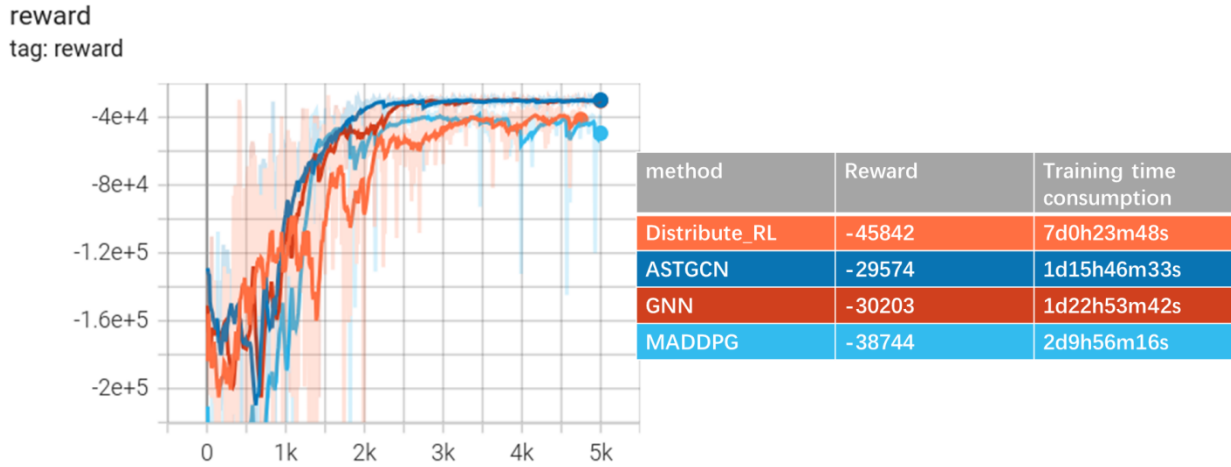


Figure 8: Convergence Analysis for Reinforcement Learning Methods

As shown in the figure 8, taking the training status of Light0 as an example, in terms of dynamic performance, the MADDPG method outperforms the distributed reinforcement learning method in convergence rate, steady-state performance, and post-convergence stability. Notably, ASTGCN and GNN introduce significant improvements to the training process, accelerating algorithm convergence and enhancing the performance ceiling, with ASTGCN demonstrating superior performance over GNN. Moreover, ASTGCN also exhibits remarkable advantages in actual training time.

Conclusion

This study proposes an ASTGCN-enhanced MADDPG framework for adaptive traffic signal control in urban road networks. By integrating the spatio-temporal feature extraction capability of ASTGCN and the multi-agent cooperative decision-making paradigm of MADDPG, the framework effectively captures complex traffic dependencies and realizes centralized training with decentralized execution. Experiments on the SUMO simulation platform demonstrate that the proposed method outperforms multiple baseline approaches in convergence speed, generalization ability, extreme traffic adaptability, and congestion reduction. This framework provides a scalable and effective data-driven solution for intelligent traffic signal control, with important application value for smart city construction and urban mobility optimization.

Discussion

The spatio-temporal attention-based architecture demonstrates strong scalability through flexible topological representations, enabling effective deployment in complex traffic networks beyond traditional four-way intersections. Furthermore, the decentralized framework realizes localized controllers via modular design, facilitating large-scale traffic network optimization while maintaining coordination efficiency.

The current limitations include: (1) discrepancies between the simulated road network and real-world traffic environments. Although actual traffic flows are adopted, the behavioral patterns of real vehicles still exhibit certain gaps compared to simulated vehicles; (2) the continued use of lane-level fairness metrics rather than vehicle-level ones. Future research will focus on enhancing policy fairness through optimized action rewards, while pursuing higher-fidelity simulation environments to bridge the gap with reality.

Data availability

The data used in this paper were all derived from the website <https://traffic-signal-control.github.io/dataset.html>. [18~20]

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Competing interests (mandatory)

The authors declare no competing interests.

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Author contributions

J.W., X.W. and C.H. conceived the experiments. X.W. and Y.M. conducted the experiments, and X.W. and J.W. performed statistical analysis and figure generation. G.L., Y. L. and H. Z. contributed to critical revision of the manuscript for important intellectual content. All authors reviewed the manuscript.

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